

# BHRUGU SETLUR

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## EDUCATION

### New York University Tandon School of Engineering

New York, NY

*B.S. Electrical and Computer Engineering | GPA: 3.74 | Dean's List: all semesters, 2025-2026*

*Expected May 2028*

- Coursework: Computer Architecture & Digital Logic (Verilog), Data Structures & Algorithms, Object-Oriented Programming (C++)
- Planned for Fall 2026: Signals & Systems, Operating Systems, Embedded Systems Design

## EXPERIENCE

### Generex

San Francisco, CA

*Embedded Firmware Engineer (Contract)*

*Jun 2026 - Present*

- Developing STM32G071 firmware for a battery monitor: 4 I2C time-of-flight sensors, reporting over an opto-isolated BACS bus.
- Separated protocol and alarm logic from the drivers into portable C, covered by 8,444 host checks that run without target hardware.
- Designed debounce, hysteresis, and confidence-over-time logic to reject false alarms from condensation, bubbles, and detached sensors.
- Validated the BACS CRC across 125,323 frames captured from Generex's official Reader and Simulator, with 0 mismatches.
- Fit the current firmware, excluding diagnostics, into about 12 KB of the chip's 36 KB of RAM.

### Tokonoma.ai

San Francisco, CA

*Software Engineering Intern*

*Jun 2026 - Aug 2026*

- Prototyped a faster vector-search path, cutting median query time 1.95x on a 20,000-chunk corpus with no measured quality loss.
- Caught a filtered-search slowdown in review and added a tested exact-search fallback plus per-query monitoring.
- Built a labeled benchmark to tune duplicate detection, scoring 81 configurations against matching and non-matching record pairs.
- Built one issue-tracking service for GitHub, GitLab, Jira, and Linear that imports issues and closes them when fixes merge.
- Reviewed the split of a 2,851-line store module, fixing clock-mismatch timeouts and stalled jobs that falsely reported healthy.
- Fixed 5 security vulnerabilities, including prompt injection in an LLM-based extraction path.
- Built passwordless email sign-in and a protected coming-soon dashboard for the company website.

Tech: Python, Rust, PostgreSQL, pgvector, Kubernetes, Helm, Docker, OpenTelemetry, Quickwit, OAuth 2.1, MCP, RAG

### Growcery | growcery.us

New York, NY

*Software Engineering Intern*

*Jan 2026 - May 2026*

- Launched a two-sided marketplace and integrated Stripe payments with Apple Pay and automatic platform-fee collection.
- Implemented radius-based seller search with Supabase and Mapbox, plus in-app messaging and seller listing tools.

### VIP: High Performance Computing at NYU

New York, NY

*Virtual Environment Penetration Tester*

*Sep 2025 - Present*

- Ran an authorized 8-phase penetration test: scanned with nmap, then cracked MD5 password hashes pulled from an open FTP share.
- Used SQL injection to dump the database, path traversal to read an SSH key, and command injection to open a reverse shell.
- Recovered credentials from an exposed .git, escalated to root via a sudo misconfiguration, then reused passwords from FTP to SSH.

## PROJECTS

### 8-bit CPU with Virtual Memory & Cache | github.com/bhrugusetlur-art/cpu-design

Jun 2026 - Aug 2026

- Designed the RTL for an 8-bit Harvard CPU: a 12-instruction ISA, ALU, register file, control unit, and datapath, backed by a write-back cache hierarchy with a direct-mapped L1 and a 4-way set-associative L2 using LRU replacement.
- Added virtual memory with a 4-entry TLB, hardware page-table walks, and page-fault handling, verified across 15 Verilog regression suites covering ISA operations, cache and LRU eviction, TLB walks, and frozen fault state, in simulation and on a Basys 3 FPGA.
- Hardened the core to GDS on SkyWater 130 nm with OpenROAD: 12,778 standard cells (3,824 sequential) at about 19% utilization, with clean DRC and antenna checks and project-deck LVS matching all 146 circuit pairs.
- Met the 10 MHz timing target with 90.29 ns worst setup slack, an estimated 103 MHz maximum, and wrote a two-pass C++ assembler that resolves labels and emits machine code for FPGA programs.
- **Open source:** Merged a regression test into **cocotb**, a widely used Python framework for RTL verification.

Tech: Verilog, RTL Design, FPGA (Xilinx Artix-7), Vivado, Icarus Verilog, cocotb, GTKWave, OpenROAD, Sky130, C++

### LineLab | NYC Transit Planning | github.com/Tandonites/LineLab

Apr 2026

- Co-developed a 5-person NYC transit sandbox simulating ridership, operating cost, and travel time for proposed corridors.
- Designed and debugged the simulation model and built the corridor-drawing interface used to run it.
- Worked across the stack: the React and Leaflet map, the FastAPI simulation API, and the NetworkX travel-time graph.
- Built radius selection and ridership heat maps; fixed GTFS routing by linking directional platform nodes to parent stations.

Tech: React, TypeScript, Vite, Leaflet, Python, FastAPI, NetworkX, MTA GTFS

## SKILLS

**Languages:** Verilog, C, C++, Python, Rust, TypeScript, SQL, Java, C#

**AI/ML:** RAG, semantic search, vector embeddings (pgvector), HNSW, LLM agents, Model Context Protocol (MCP)

**Frameworks & Tools:** React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Kubernetes, Helm, Docker, OpenTelemetry, Git, GCP, Vercel, Nmap, sqlmap

**Hardware & Verification:** RTL Design, ASIC Physical Design (OpenROAD, Sky130), STA, DRC, LVS, FPGA (Xilinx Artix-7), Vivado, Icarus Verilog, cocotb, GTKWave, STM32 (ARM Cortex-M0+), I2C, UART

## AWARDS AND HONORS

**3rd Place, Code4City Hackathon** | 6th Place, USA Fencing Div. II Nationals